

AVIRAL TRIPATHI

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EDUCATION

KLE Technological University, Hubballi, Karnataka

Aug 2024 – Present

Bachelor of Engineering in Computer Science CGPA: 8.34

CodeClub KLETU – PR member

Defence Club KLETU – Content Creation Lead

PROJECT EXPERIENCE

- **ResQRoute – Emergency Coordination Platform** [Live](#) / [GitHub](#) May 2026
Next.js 15, TypeScript, PostgreSQL, Prisma, Tailwind CSS
Built a real-time hotel emergency management system coordinating guests, staff, and first responders during fire, medical, and structural incidents.
 - Implemented a Dijkstra-driven evacuation pathfinder over a Prisma graph, applying dynamic distance and floor-penalty weighting for real-time hazard-aware rerouting.
 - Developed a multilingual speech-to-triage pipeline using AssemblyAI + OpenRouter LLM classification, generating structured JSON dispatch reports with **90%+ intent accuracy** across emergency categories.
 - Built an automated emergency dispatch system using Twilio SMS/Voice with room-level location tracking, plus role-based Next.js dashboards backed by Prisma + Supabase Postgres, enabling **sub-3s dispatch alerts** and supporting **50+ concurrent emergency events** in stress tests.
- **PulsePoint - AI Family Health Network** [Live](#) / [GitHub](#) Apr 2026
Python, FastAPI, Pydantic v2, LightGBM, Gemini Flash, Llama 3.1 LoRA, RAG, Hugging Face Spaces (Docker)
Built a safety-first AI/ML clinical triage+diagnostics engine using a multi-model pipeline (Triage+Predict+Connect) with deterministic medical grounding.
 - **Trained** a 6-stage triage pipeline combining YAML vital-rule safety rails, **LightGBM multiclass severity classification (SHAP)**, and RAG-based WHO/MOHFW guideline retrieval for grounded clinical justification, achieving **~91–93% triage accuracy** with explainable SHAP-based reasoning.
 - **Fine-tuned Llama 3.1 using LoRA** for structured symptom interviewing, and implemented hallucination guards to block fabricated ICD-10 codes/lab tests via KB validation (LLM can escalate risk but never downgrade).
 - Built Predict + Connect modules: developed a symptom-to-ICD-10 top-K classifier, extracted 40+ lab tests via OCR/regex with LLM explanations, and ranked nearby doctors using ICD-10 specialty matching and distance-based urgency sorting.
- **Streamly - YouTube Clone** [GitHub](#) Jan 2026
Node.js, Express.js, MongoDB, Mongoose, Cloudinary
 - Developed a scalable video streaming backend with RESTful APIs for authentication, video uploads, and metadata management, including optimized video discovery using MongoDB aggregation and pagination.
 - Implemented secure JWT-based authentication (access+refresh tokens) with bcrypt hashing and cookie-based sessions, and built a media upload pipeline using Multer and Cloudinary.

TECHNICAL SKILLS

Languages: C/C++, JavaScript, TypeScript, Python, SQL

Core Concepts: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Frameworks/Libraries: React.js, Next.js, Express.js, Node.js, Tailwind CSS, HTML, CSS

AI/ML: LightGBM, SHAP, LoRA Fine-Tuning, RAG (ChromaDB, pgvector), LLM APIs (Gemini, OpenRouter)

Databases: MongoDB, PostgreSQL, MySQL

Tools/Concepts: Git, GitHub, Docker, Postman, VS Code, Vercel

ACHIEVEMENTS

- **Winner – Hack Karnataka (HealthTech Track)** – 1st place among 50+ finalist teams at a national-level hackathon.
- **Finalist – AlgoBlitz:** Achieved a Top 3 ranking in a university-wide Data Structures and Algorithms Competition.
- **Top 10 – QuantumArena 1.0 & Cepheus 2.0:** Recognized among top teams in competitive hackathons.
- **Solved 600+ problems on Codeforces and LeetCode** with regular contest participation.